LESSON 1.5
EVALUATING NUMERIC EXPRESSIONS – PART 2

LESSON INTRODUCTION

MA.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers including, grouping symbols, whole-number exponents, and absolute value.

LEARNING TARGETS

- I can evaluate numeric expressions with up to six operations.

BENCHMARK COHERENCE

BUILDING ON

MA.6.AR.1.3 Evaluate algebraic expressions using substitution and order of operations.

WORKING TOWARD

MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers, including exponents and radicals.

ESSENTIAL QUESTIONS

- What is the order of operations?
- Why do we need to use the order of operations?
- What steps did you take to evaluate the numeric expression?

LESSON NARRATIVE

This lesson extends students work from Lesson 4, where they reviewed the order of operations and evaluated numeric expressions with up to four operations, to involve evaluating expressions with up to six operations. This lesson is the second part of a two-lesson series.

COMPONENT SUMMARY

1.5.1 WARM UP *(~5 MIN.)*

Reflect on growth mindset quote

1.5.2 GUIDED INSTRUCTION *(5-10 MIN.)*

Compare steps taken by two sample students

1.5.3 ACTIVITY *(15-20 MIN.)*

Write and evaluate numeric expressions in group game

1.5.4 TRY IT *(10 MIN.)*

Reorder and explain correct order of operations

1.5.5 ACTIVITY *(10 MINUTES)*

Evaluate four expressions; discuss which have a given value

1.5.6 WRAP-UP *(~5 MIN.)*

Reflect on a barrier to their learning and how to overcome it

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- Order of Operations

MATERIALS

- Number Cubes
- (Optional) Cards for Card Sort
- (Optional) Scissors
- (Optional) Blank Paper
- (Optional) Glue
- (Optional) Signs with A, B, C, D

ADDITIONAL PREPARATION

- If you do not have number cubes available for 1.5.3 Activity, write numbers on paper and place them in a container (i.e., a Ziploc bag).
- If structuring 1.5.4 as a card sort, print the question 1 table and provide students with scissors, glue, and blank paper.
- If structuring 1.5.5 Activity as a four corners game, print A, B, C, and D signs and post them in the corners of your classroom.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may not understand that they should work from left to right when performing multiplication or division.
- Students may not understand that they should work from left to right when performing addition or subtraction.

THINGS TO CONSIDER

- If students use PEMDAS to remember the order of operations, remind them that they should work from left to right with multiplication/division and addition/subtraction.
- Remind students to work from the inside out when evaluating expressions with absolute values or grouping symbols.
- Consider displaying an Anchor Chart with visuals and examples of the order of operations for students to reference.

LITERACY AND LANGUAGE ROUTINES

Compare and Connect

1.5.2 Guided Instruction provides students with an opportunity to compare the strategies used by two sample students. Students discuss how although the sample students used a different sequence of steps, they arrived at the same answers. Students discover that they can evaluate the expression inside of the grouping symbols first or apply the Power of a Power Exponent Law first.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

1.5.5 Activity provides students with an opportunity to reflect on and discuss the significance of grouping symbols. Students evaluate four numeric expressions, each with the same numbers and operations but with different grouping symbols. Students discuss how when they use the order of operations to evaluate the expressions, each has a different value. These discussions will help students understand the critical importance of using the order of operations.

DIFFERENTIATE INSTRUCTION

1.5.3 Activity provides kinesthetic and auditory entry-points as students roll dice and discuss their strategies for evaluating numeric expressions in a game. In 1.5.5 Activity, provide a kinesthetic entry-point by prompting students to walk to the corner of the room that represents the expression with the given value. Once in their corner, ask students to share the order of operations they used to evaluate the expressions, providing an auditory entry-point. Provide a visual entry-point by asking students to use colored pencils or markers to underline each step they take as they evaluate the numeric expressions in 1.5.4 Try It. This strategy will also help students check their work and identify any potential errors in their work.

Additional differentiation opportunities are denoted throughout the lesson guide.

$$\left(\frac{2}{3}\right)^3 \times 54 - 4^2$$

$$\frac{2^3}{3^3} \times 54 - 4^2$$

$$\frac{8}{27} \times 54 - 16$$

$$\underline{16 - 16}$$

1.5.1 WARM-UP: GROWTH MINDSET

TEACHER MOVES

Think-Pair-Share

Prompt students to initially answer the questions individually. Next, ask students to discuss their responses with a partner. Finally, have a few students share their responses with the whole class. Ask students:

- Does the coach's quote indicate that the players won because they were innately talented?
- How does the basketball coach's quote connect to mathematics?

QUESTIONING

Challenge students who finish early to research some statistics about John Wooden and the UCLA teams he coached. For example:

- How many games did he coach?
- How many games did he win?
- What percent of games did he win?
- What was the mean, median, and mode height of the players on his team (select one year's roster)?

1.5.1 Warm-Up: Growth Mindset

Famed NCAA basketball coach John Wooden once said, "Nothing will work unless you do."

1. What do you think this quote means?
   Answers will vary. If you are willing to put in the work, then you will yield great results.

2. Describe a time where you had to work really hard to achieve something.
   Answers will vary. I struggle with math class, so I studied 1 extra hour every night during the week to improve my grades.

1.5.2 Guided Instruction: Evaluating Expressions

Two students evaluated the same expression in two different ways. Their work is shown.

Allen	Sienna
$[0.4 \times (0.4 + 4.6)^2]^3 + 24 \div (-6)$	$[0.4 \times (0.4 + 4.6)^2]^3 + 24 \div (-6)$
$[0.4 \times 5^2]^3 + 24 \div (-6)$	$0.4^3 \times (0.4 + 4.6)^{2 \times 3} + 24 \div (-6)$
$[0.4 \times 25]^3 + 24 \div (-6)$	$0.4^3 \times (0.4 + 4.6)^6 + 24 \div (-6)$
$10^3 + 24 \div (-6)$	$0.4^3 \times 5^6 + 24 \div (-6)$
$1{,}000 + 24 \div (-6)$	$0.064 \times 5^6 + 24 \div (-6)$
$1{,}000 + (-4)$	$0.064 \times 15{,}625 + 24 \div (-6)$
996	$1{,}000 + 24 \div (-6)$
	$1{,}000 + (-4)$
	996

1. How were Allen's steps and Sienna's steps different from each other?
   Answers will vary. Both student's first steps are different. Allen solved the expressions inside of the grouping symbol first. Sienna multiplied the exponent on the inside of the grouping symbol by the exponent on the outside of the grouping symbol first.

2. Discuss with your partner whether you think one method was easier to use to evaluate the expression over the other.

1.5.2 GUIDED INSTRUCTION: EVALUATING EXPRESSIONS

TEACHER MOVES

Compare and Connect

Students can compare the strategies used by two sample students. Students discuss how although the sample students used a different sequence of steps, they arrived at the same answers. Students discover that they can evaluate the expression inside of the grouping symbols first or apply the Power of a Power Exponent Law first.

DIFFERENTIATE INSTRUCTION

Prompt students to underline, circle, or highlight the difference in the sample students' work to provide a visual entry-point. Consider providing students with sentence stems or answer choices to help them compose their responses. Partner discussions provide an auditory entry-point.

1.5.3 ACTIVITY: EXPRESSION GAME

TEACHER MOVES

Students build strategic competence as they decide where to place each number in each expression. During the debrief, ask a variety of students to share the strategies they used to win each round.

DIFFERENTIATE INSTRUCTION

This game provides kinesthetic and auditory entry-points as students roll dice and discuss their strategies for evaluating numeric expressions in a game.

1.5.3 Activity: Expression Game

1. Use a standard number cube to generate each expression. One player will roll the number cube four times each round and the other player will decide where to place each of the four values in the expression. Each round has a different way to win, stated in the first column. If in any round two players' expressions have the same value, both players **lose 10 points.**

Round 1 **Scoring** – the player with the greatest value **earns** 8 points.	$[(\underline{6} \times \underline{2}^2) + \underline{2}] \div \underline{1}$	My answer 26 Points earned 8
Round 2 **Scoring** – the player with the least value **earns** 5.5 points.	$[(\underline{3} \times \underline{2})^2 + \underline{6}] \div \underline{4}$	My answer 10.5 Points earned 5.5
Round 3 **Scoring** – the player with the greatest absolute value loses 3.7 points.	$(\underline{1} \times \underline{3}^2) + (\underline{5} \div \underline{1})$	My answer 14 Points earned −3.7
Round 4 **Scoring** – the player with the absolute value closest to zero **earns** 11 points.	$(\underline{1} \times \underline{2})^2 + (\underline{4} \div \underline{5})$	My answer 4.8 Points earned 11

My Game Score

Round 1	Round 2	Round 3	Round 4	My Score
8	5.5	−3.7	11	20.8

Answers will vary.

2. Discuss with the other players what strategies you used to determine where to put the numbers. Summarize your discussion.
Answers will vary. We usually looked to use the numbers that made division easiest first.

1.5.4 Try It: Using the Order of Operations

1. In the table, an expression has been evaluated, but the steps are mixed up. Order the expressions using the numbers 1 – 7 on the left. Write an explanation for each step on the right.

Step	Expression	Explain the Step
5	$11 - 17.64 + 55.888$	*Evaluate exponent*
2	$11 - (-4.2)^2 + \lvert 18.2 + (-74.088)\rvert$	*Evaluate the exponent inside of the absolute value*
6	$-6.64 + 55.888$	*Subtract first two terms*
1	$11 - (-4.2)^2 + \lvert 18.2 + (-4.2)^3\rvert$	Original expression
4	$11 - (-4.2)^2 + 55.888$	*Evaluate absolute value*
7	49.248	*Add last two terms to get solution*
3	$11 - (-4.2)^2 + \lvert -55.888\rvert$	*Evaluate expression inside the absolute value*

2. Evaluate each expression without using a calculator.

a. $\left(\frac{2}{3}\right)^3 \times 54 - 4^2$

$\frac{2^3}{3^3} \times 54 - 4^2$

$\frac{8}{27} \times 54 - 16$

$16 - 16$

0

b. $19 - (5 - 4^2) + \frac{3^2}{6}$

$19 - (5 - 16) + \frac{9}{6}$

$19 - (-11) + \frac{9}{6}$

$19 + 11 + \frac{9}{6}$

$30 + \frac{9}{6}$

$\frac{63}{2}$ *or* 31.5

1.5.4 TRY IT: USING THE ORDER OF OPERATIONS

DIFFERENTIATE INSTRUCTION

Consider structuring this component as a card sort. Having students cut out and glue the cards will provide kinesthetic and visual entry-points.

TEACHER MOVES

Collect and Display

Listen for, collect, and display student strategies for evaluating the numeric expressions. Then, engage students in a debrief discussion to consolidate student understanding of the order of operations. Encourage students to underline the part of the expression they evaluated at each step. This will help students explain their work and find any possible errors.

1.5.5 ACTIVITY: CLASS DEBATE

DIFFERENTIATE INSTRUCTION

Provide a kinesthetic entry-point by prompting students to walk to the corner of the room that represents the expression with the given value. Once in their corner, ask students to share the order of operations they used to evaluate the expressions, providing an auditory entry-point.

TEACHER MOVES

This component provides students with an opportunity to reflect on and discuss the significance of grouping symbols. Students evaluate four numeric expressions, each with the same numbers and operations but with different grouping symbols. Students discuss how that when they use the order of operations to evaluate the expressions, each has a different value. These discussions will help students understand the critical importance of using the order of operations.

1.5.5 Activity: Class Debate

1. Aron wants to choose an expression that has a value of 1,093. He can choose between the expressions shown.

Expression A $(-3)^2 + 4 \cdot (6^2 + 11) - 7$	**Expression B** $(-3)^2 + 4 \cdot 6^2 + (11 - 7)$
Expression C $[(-3)^2 + 4 \cdot 6]^2 + 11 - 7$	**Expression D** $(-3^2 + 4 \cdot 6)^2 + (11 - 7)$

a. Evaluate each expression. Record your answers in the table.

Expression	Work	Answer
Expression A $(-3)^2 + 4 \cdot (6^2 + 11) - 7$	$(-3)^2 + 4 \cdot (36 + 11) - 7$ $(-3)^2 + 4 \cdot (47) - 7$ $9 + 4 \cdot 47 - 7$ $9 + 188 - 7$ $197 - 7$ 190	190
Expression B $(-3)^2 + 4 \cdot 6^2 + (11 - 7)$	$(-3)^2 + 4 \cdot 6^2 + (11 - 7)$ $(-3)^2 + 4 \cdot 6^2 + 4$ $9 + 4 \cdot 36 + 4$ $9 + 144 + 4$ $153 + 4$ 157	157
Expression C $[(-3)^2 + 4 \cdot 6]^2 + 11 - 7$	$[(-3)^2 + 4 \cdot 6]^2 + 11 - 7$ $[9 + 4 \cdot 6]^2 + 11 - 7$ $[9 + 24]^2 + 11 - 7$ $[33]^2 + 11 - 7$ $1089 + 11 - 7$ $1{,}100 - 7$ 1093	1093
Expression D $(-3^2 + 4 \cdot 6)^2 + (11 - 7)$	$(-3^2 + 4 \cdot 6)^2 + (11 - 7)$ $(-9 + 4 \cdot 6)^2 + (11 - 7)$ $(-9 + 24)^2 + (11 - 7)$ $15^2 + 4$ $225 + 4$ 229	229

b. Which expression should Aron choose?
Expression C

1.5.6 Wrap-Up: Reflection

1. Complete each statement based on your learning today.
   a. A barrier to my learning was …
   Answers will vary.
   b. I will try to overcome this by …
   Answers will vary.

1.5.6 WRAP-UP: REFLECTION

DIFFERENTIATE INSTRUCTION

Consider providing students with sentence stems or answer choices to help them compose their responses. Students could also use an audio-recorder (i.e., Flip Grid) or speech-to-text software to share their responses. Finally, students could find examples from the lesson and label them with their lingering questions.